Level 1 Functional Skills Mathematics
SAMPLE PAPER 3

City & Guilds
A City & Guilds Group Business

Version: 1.0

Duration: 25 minutes
Total marks: 15

MODEL SOLUTION SCORE
15 / 15 marks (100% Pass)
25 minutes · Verified by MFS

SECTION 1 – CALCULATOR NOT PERMITTED

Candidate name (first, last)

First: MASTER

Last: FUNCTIONAL SKILLS

Candidate enrolment number: MFS4748

Date of birth (DDMMYYYY): 01012000

Assessment date (DDMMYYYY): 20102026

Centre number: MFS0001

Candidate signature and declaration*: Master Functional Skills

- If additional answer sheets are used, enter the additional number of pages in this box.
- Before taking the examination, **all candidates** must check that their barcode label is in the appropriate box. Incorrectly placed barcodes may cause delays in the marking process.
- Please ensure that you staple additional answer sheets to the **back** of this answer booklet, clearly labelling these with your full name, enrolment number, centre number and qualification number in BLOCK CAPITALS.
- All candidates need to use a **black/blue** pen. **Do not** use a pencil or gel pen, unless otherwise instructed.
- If provided with source documents, these documents **must** be returned to City & Guilds. Do not write on the source documents.

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Please check that your name is correctly printed on the candidate barcode label. If not, please tell the invigilator before the start of the exam.

You should have the following for this assessment:

- a pen with black or blue ink
- a pencil (for diagrams, graphs and charts only)
- an eraser
- a 30cm ruler
- a protractor.



You must NOT use a calculator for Section 1.

General instructions

- Read through each question carefully.
- You may use a dictionary.
- Write all your answers in this booklet.
- Check your calculations and check that your answers make sense.

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SECTION 1 – CALCULATOR NOT PERMITTED

There are **15** marks available in this section.

You should check all your work as you go along.

You must **not** use a calculator in this section.



Q1

4.7 x 1000 =

• Multiply by 1,000: move decimal 3 places right --> 4,700

4 700

[1 mark]

(1 mark)

Q2

16 ⅔ is the same as

(tick one box)

A $\frac{18}{3}$ ☐

B $\frac{50}{3}$ ☑

• 16 2/3 = (16 x 3 + 2) / 3 = (48 + 2) / 3 = 50/3

[1 mark]

C $\frac{32}{3}$ ☐

D $\frac{48}{3}$ ☐

(1 mark)

Q3

What is 8% as a decimal?

• 8% = 8 ÷ 100 = 0.08 (divide by 100)

0.08

[1 mark]

(1 mark)

Q4

12^2 =

• $12^2 = 12 \times 12 = 144$

144

[1 mark]

(1 mark)

Q5

What is 7.2857 rounded to two decimal places?

- 7.2857: 3rd decimal place is 5, so round up 8 to 9 --> 7.29

7.29

[1 mark]

(1 mark)

Q6

$8 + (7 - 5) \times 3 =$

- BIDMAS: Brackets (7 – 5) = 2 --> Multiply 2 x 3 = 6 --> 8 + 6 = 14

14

[1 mark]

(1 mark)

Q7

What is two hours and thirty eight minutes in **minutes**?

- 2 hours = 2 x 60 = 120 minutes --> 120 + 38 = 158 minutes

158

[1 mark]

minutes

(1 mark)

Q8

What is $\frac{2}{5}$ of 300m?

- (300 ÷ 5) x 2 = 60 x 2 = 120 m

120

[1 mark]

metres

(1 mark)

Q9

Which one of these lists of decimals is in **increasing** order?

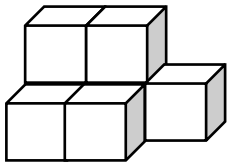
(tick one box)

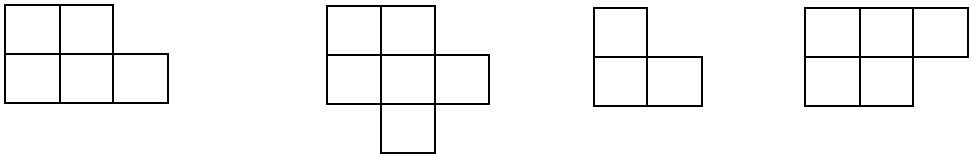
A	0.401	0.42	0.503	0.59	☑
B	0.59	0.503	0.42	0.401	☐
C	0.42	0.59	0.401	0.503	☐
D	0.503	0.59	0.42	0.401	☐

• 0.401, 0.420, 0.503, 0.590 (incre [1 mark]

Q10

Which is a **plan view** of the object shown?





(tick one)

A ☐ **B** ☐ **C** ☐ **D** ☑

• Plan view is from directly above: shows full 2D footprint (Option D)

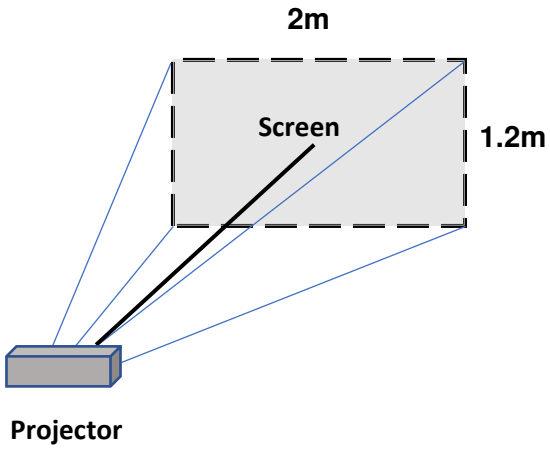
[1 mark] **(1 mark)**

Q11

A community centre projects films on to a screen.

The manager wants to make a wooden border around the screen.

She needs to order the wood.



_ _ _ _ Border

What length of wood does she need to order?

Show your working

- Wooden border around rectangular screen = Perimeter of rectangle
- Perimeter = 2 m + 1.2 m + 2 m + 1.2 m = 6.4 m [M1]

(Or: 2 × (2 + 1.2) = 2 × 3.2 = 6.4 m)

6.4 m

[2 marks]

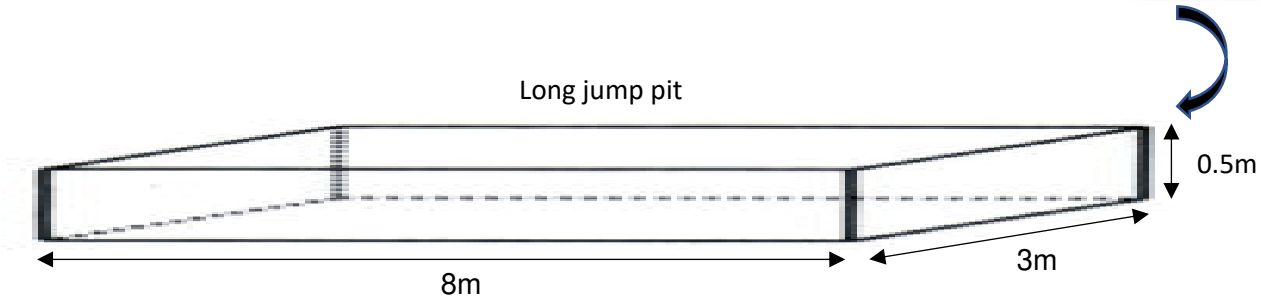
(2 marks)

Q12

A sports club needs to replace the sand in a long jump pit.

The groundsman works out the capacity of the long jump pit to order enough sand.





What amount of sand does he order?
Put units on your answer.

Show your working

- Capacity (Volume) = Length × Width × Depth [M1]
- Volume = 8 m × 3 m × 0.5 m = 24 × 0.5 = 12 m³ [A1]
- State correct cubic units: m³ (or cubic metres) [B1]

12 m³

[3 marks]

(3 marks)

Level 1 Functional Skills Mathematics
SAMPLE PAPER 3

A City & Guilds Group Business

Version: 1.1

Duration: 1 hour 20 minutes
Total marks: 45

MODEL SOLUTION SCORE
45 / 45 marks (100% Pass)
1 hr 20 mins · Verified by MFS

SECTION 2 – CALCULATOR PERMITTED

Candidate name (first, last)

First: MASTER

Last: FUNCTIONAL SKILLS

Candidate enrolment number: MFS4748

Date of birth (DDMMYYYY): 01012000

Assessment date (DDMMYYYY): 20102026

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General instructions

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- You may use a dictionary.
- Show your working out (where required).
- Write all your working out and answers in this booklet.
- Check your calculations and check that your answers make sense.
- There are additional pages **including graph paper** at the back of this booklet if you run out of space or ask the invigilator if you need additional sheets of paper.

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SECTION 2 – CALCULATOR PERMITTED

There are **45** marks available in this section.

You should check all your work as you go along.

You may use a calculator.



Q1

What is 10% of £68.50?

- 10% = 1/10 --> £68.50 ÷ 10 = £6.85

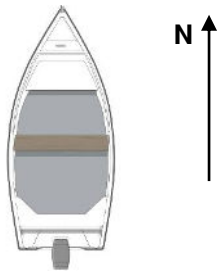
£ 6.85

[1 mark]

(1 mark)

Q2

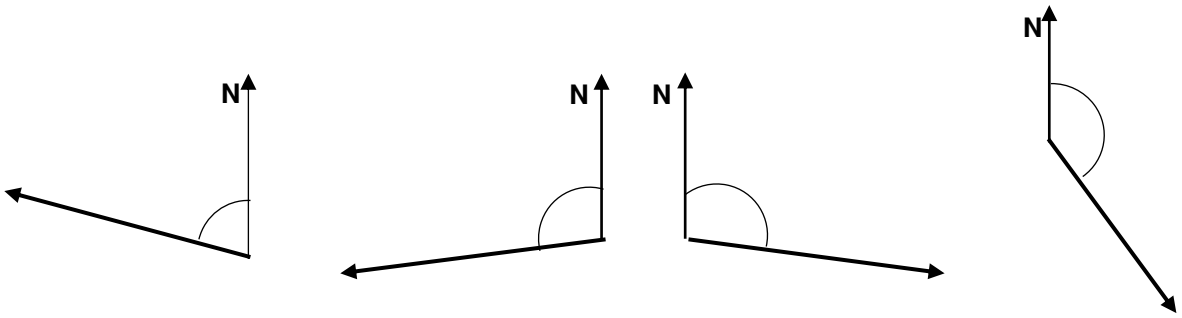
A boat is sailing North.



It turns 100 degrees in an **anti-clockwise** direction.

Which one of the following shows the direction in which the boat is now sailing?

(tick one box)



A ☐ B ☑ C ☐ D ☐

- Starting North (0°/360°), turn anti-clockwise 100°: 360° − 100° = 260°
- West is 270°, so 260° points just south of west (Diagram B)

[1 mark]

(1 mark)

Q3

A sea diver makes 7 different dives.

He measures how deep he dives.

This table shows his results:

Dive Number	Depth
1	-14m
2	-21m
3	-7m
4	-25m
5	-19m
6	-32m
7	-18m

What was the variation in depth for the seven dives?

Show your working

- Variation = Range = Deepest dive – Shallowest dive
- Deepest = 32 m (or –32 m); Shallowest = 7 m (or –7 m)
- Variation = 32 m – 7 m = 25 metres [B1]

Variation 25 **metres**

[1 mark]

(1 mark)

Q4

What is the interest paid on £2500 at an interest rate of 15%

- 15% of £2,500 = £2,500 × 0.15 = £375 (or £250 + £125)

£375

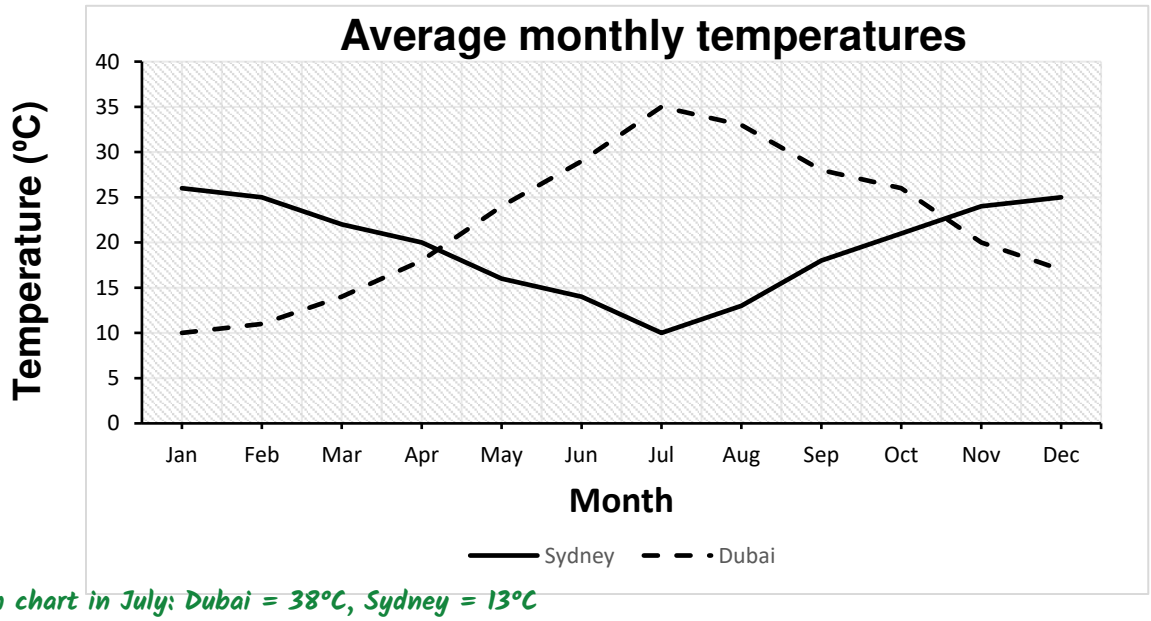
[1 mark]

(1 mark)

Q5

This chart shows the average monthly temperatures of Sydney and Dubai.

What is the difference in the average monthly temperature of Sydney and Dubai in July?



- From chart in July: Dubai = 38°C, Sydney = 13°C
- Difference = 38°C − 13°C = 25°C

25 °C

[1 mark]

(1 mark)

Q6

An aircraft passenger weighs his bag.

He goes to the check-in desks and says that his bag weighs 14g.

The person at the desk says this must be incorrect.

Explain why.

- 14 grams is far too light for a suitcase (about the weight of a pen!).
- He has confused grams with kilograms; the bag weighs 14 kg, not 14 g.

Explanation

14g is the wrong order of magnitude for luggage weight. He has mistaken grams (g) for kilograms (kg). An aircraft bag weighs approximately 14 kg.

[1 mark]

(1 mark)

Q7

A parent wants a car seat for his child who is 36 months old.

He needs a seat that is front facing and uses a harness.

He looks at this table.

Car Seats					
		Features			
Child's age	**Seat type**	**Lie Flat**	**Front Facing**	**Rear Facing**	**Uses harness**
From birth - 12 months	Baby	✓	X	✓	✓
6 months – 4 years	Toddler A	X	X	✓	✓
	Toddler B	X	✓	✓	✓
3.5 years – 12 years	Child A	X	✓	✓	✓
	Child B	X	✓	X	X

Which seat type should the parent buy?

Explain your decision.

Which seat type should the parent buy?

Toddler B

Explanation

1. 36 months = 36 ÷ 12 = 3 years old, which falls in the range 6 months – 4 years.

2. Looking at the table, Toddler B is both front facing AND uses a harness,

whereas Toddler A is rear facing. Therefore, Toddler B meets all requirements. [A1]

[2 marks] **(2 marks)**

Q8

Dave and Sue are community centre workers.

They share the opening of the centre each day.

Their manager shares this work out between Dave and Sue in the ratio 3:1

She makes this rota for the first 16 days of the month.

	Name		Name
Day 1	Dave	Day 9	Dave
Day 2	Dave	Day 10	Sue
Day 3	Dave	Day 11	Dave
Day 4	Dave	Day 12	Dave
Day 5	Dave	Day 13	Sue
Day 6	Dave	Day 14	Sue
Day 7	Dave	Day 15	Sue
Day 8	Dave	Day 16	Dave

She needs to complete the rota for the next 5 days.

Complete the rota

Show your working

- Ratio of shifts Dave : Sue = 3 : 1 --> Total 4 parts
- 16 days ÷ 4 parts = 4 days per part
- Dave required: 3 × 4 = 12 days [M1] | Sue required: 1 × 4 = 4 days [M1]
- In Days 1–11: Dave has worked 10 days, Sue has worked 1 day
- Remaining 5 days: Dave needs 2 more days (12 − 10), Sue needs 3 more days (4 − 1) [A1]

[3 marks]

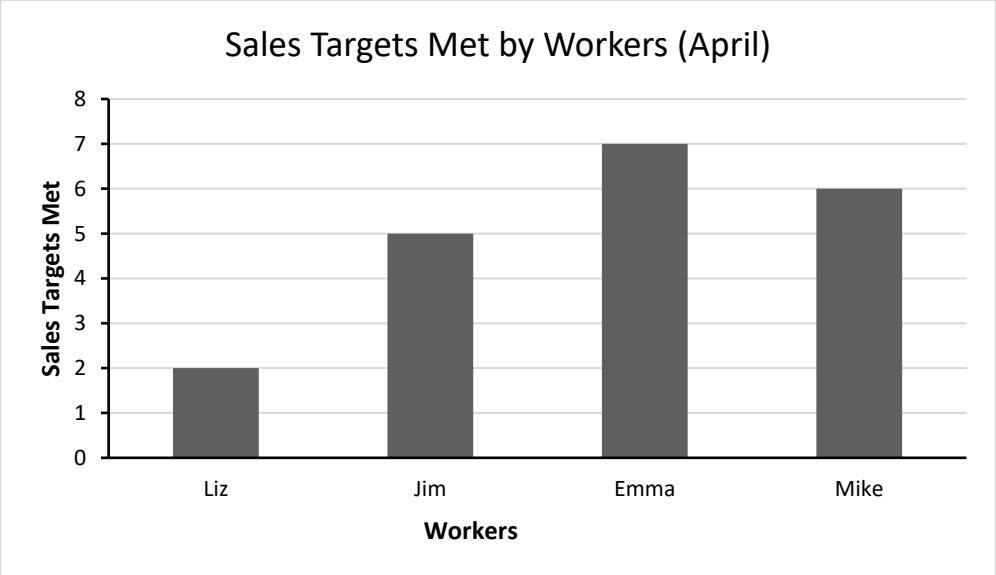
(3 marks)

Q9

A workplace gives one raffle ticket to its workers for each sales target that they meet.

Each month one ticket is drawn at random for a prize.

The chart shows the number of sales targets met by each worker.

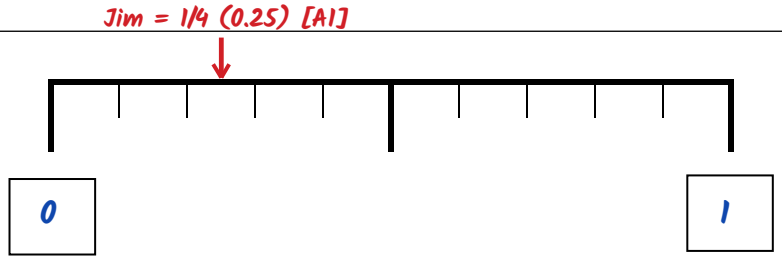


What is the probability that Jim will win the April draw?

Complete the probability diagram below to show your answer.

Show your working

- Total tickets = 6 (Liz) + 5 (Jim) + 7 (Emma) + 2 (Mike) = 20 tickets [M1]
- Jim's tickets = 5 --> Probability = 5/20 = 1/4 = 0.25 (or 25%) [A1]



[4 marks]

(4 marks)

Q10

An inspector in a crisps factory measured the weight of crisps in 20 bags to check the accuracy of their weights.

The bags were labelled **Average Contents 25g**

The table below shows the weight of the bags of crisps.

Weight of bags of crisps in grams				
24.4	25.5	25.0	25.3	24.4
26.5	25.2	24.9	24.3	25.1
25.3	25.1	25.1	24.2	24.5
24.7	24.5	26.0	25.4	25.2

Present these weights in **3** suitable groups.

Make one comment about what the results show you.

Show your working

Weight Group (g)	Raw Weights	Frequency
24.0 – 24.9	24.2, 24.3, 24.4, 24.4, 24.5, 24.5, 24.7, 24.9	8
25.0 – 25.9	25.0, 25.1, 25.1, 25.1, 25.2, 25.2, 25.3, 25.3, 25.4, 25.5	10
26.0 – 26.9	26.0, 26.5	2

Total: 20 bags [M2, A1]

Comment

- Half of the bags (10 out of 20, 50%) weighed between 25.0g and 25.9g, and 12 out of 20 bags (60%) met or exceeded the advertised 25g average contents. [A1]

[4 marks]

(4 marks)

Q11

A woman is training for a triathlon race, which includes a 1.5km swim.

She needs her swim time to be 33 minutes.

She trains in a swimming pool, which is 50m long.

She needs to know the average time she must swim each length.

What is this time?

Show your working

Step 1: Convert swim distance to metres: 1.5 km = 1.5 × 1,000 = 1,500 m [M1]

Step 2: Calculate number of lengths: 1,500 m ÷ 50 m = 30 lengths [M1]

Step 3: Calculate average time per length:

33 minutes ÷ 30 = 1.1 minutes per length [M1]

Step 4: Convert 1.1 minutes to minutes and seconds:

0.1 minute = 0.1 × 60 seconds = 6 seconds --> 1 minute 6 seconds [A1]

1 minutes 6 seconds

[4 marks]

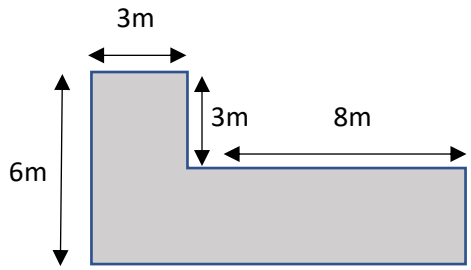
(4 marks)

Q12

A man wants to buy bags of gravel to cover his driveway.

He decides to work out the area of his driveway.

1 bag of gravel covers $14m^2$



Sketch of driveway

Not to scale

What is the area of his driveway?

How many bags of gravel must he buy?

Show your working

Step 1: Split driveway into two rectangles (Compound Shape):

- *Rectangle A: 3 m × 3 m = 9 m² [M1]*
- *Rectangle B: (8 m + 3 m) × 3 m = 11 m × 3 m = 33 m² [M1]*

Step 2: Total Area of Driveway = 9 m² + 33 m² = 42 m² [A1]

Step 3: Calculate number of bags (1 bag covers 14 m²):

42 m² ÷ 14 m² per bag = 3 bags exactly [A1]

Area of driveway 42 **m²**

Number of bags of gravel 3

[4 marks]

(4 marks)

Q13

A football manager needs to pick his team for the next match.

He gets some information on players in the team from previous matches.

He needs to choose between two players based on their work rate.

He measures their work rate using **average** and **variation** of the distance they covered.

	Distance covered for previous matches (km)						
Player A	7.2	6.8	7.6	8.7	7.4	6.8	8.0
Player B	7.9	7.3	5.8	7.6	***DNP**	***DNP**	6.9

***DNP = Did Not Play**

Which player should the manager choose?

Make two comments with reference to average distance and variation in distances.

Space for working

Player A (7 matches):

- *Total distance = 7.2 + 6.8 + 7.6 + 8.7 + 7.4 + 6.8 + 8.0 = 52.5 km*
- *Mean = 52.5 ÷ 7 = 7.5 km [M1] | Range = 8.7 − 6.8 = 1.9 km [A1]*

Player B (5 matches, 2 DNP):

- *Total distance = 7.9 + 7.3 + 5.8 + 7.6 + 6.9 = 35.5 km*
- *Mean = 35.5 ÷ 5 = 7.1 km [M1] | Range = 7.9 − 5.8 = 2.1 km [A1]*

Which player should the manager choose?

(tick one box)

Player A ✓ **Player B**

Comment about average

- *Player A has a higher average (mean) distance covered than Player B (7.5 km vs 7.1 km), showing a greater overall work rate. [A1]*

Comment about variation

- *Player A has a smaller range (variation) in distance covered than Player B (1.9 km vs 2.1 km), showing more consistent match performances. [A1]*

[6 marks] **(6 marks)**

Q14

A **father and his child** want to take self-defence lessons at a karate club for a year.

The club offers lessons for members and non-members.

The club holds one lesson per week.

Everyone who comes to lessons must buy a uniform and take out insurance for the year.

The father and child can only go to half of the lessons for the year.

Karate Club pricelist

- Membership Adult: £20 per month
- Child: £16 per month
- Non-members £8.50 per lesson **(all ages)**
- Insurance £15 per year
- Uniform £95 **(20% discount for members)**

He needs to decide whether to take lessons as members or non-members.

What should he decide to do? Explain your answer using figures.

Show your working

- 52 weeks ÷ 2 = 26 lessons attended by each person (father & child)
- Members: Fees £432 (12×£20 + 12×£16) + Ins £30 + Uniforms £152 (2×£76) = £614 [M2, A1]
- Non-Members: Lessons £442 (52 × £8.50) + Ins £30 + Uniforms £190 (2×£95) = £662 [M2, A1]
- Difference: £662 – £614 = £48 saving by joining as members [A1]

Decision *(tick one box)*

Member ☑ **Non-member** ☐

Explanation

1. He should decide to join as members because the total yearly cost for both is £614, compared to £662 as non-members.
2. Being members saves £48 over the year (£662 – £614 = £48 saving), even though they only attend half the lessons (26 weeks). Therefore, membership is cheaper. [A1]

[6 marks]

(6 marks)

Q15

The owner of a trampoline park has a floor plan.

She wants to put 4 trampolines in **zone 3**.

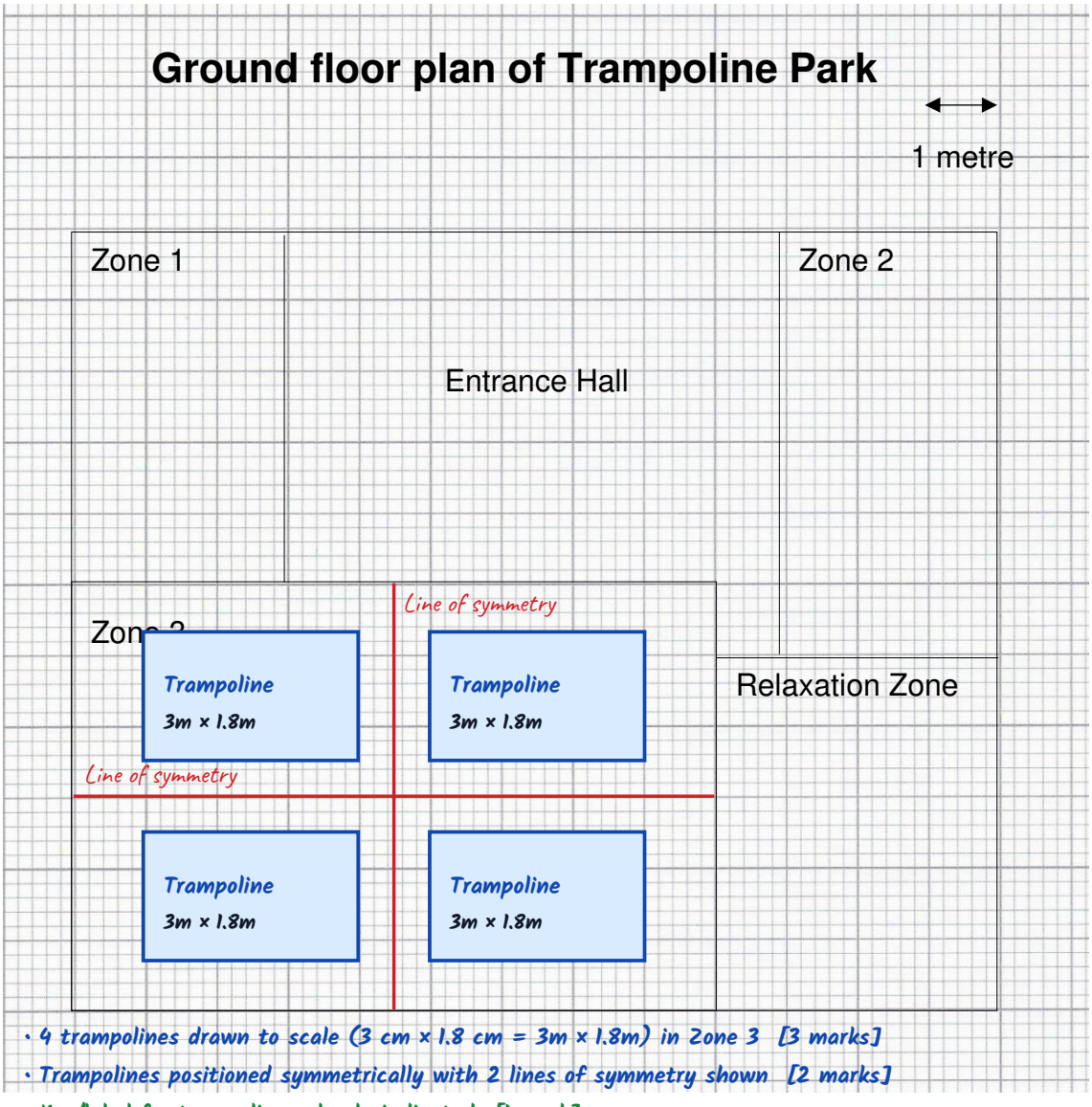
She needs a scale diagram to see where the trampolines will go.

Each trampoline measures 3 metres by 1.8 metres.

She wants them placed with 2 lines of symmetry in **zone 3**.

Complete the plan below to show the positions of the trampolines.

Draw the 2 lines of symmetry.

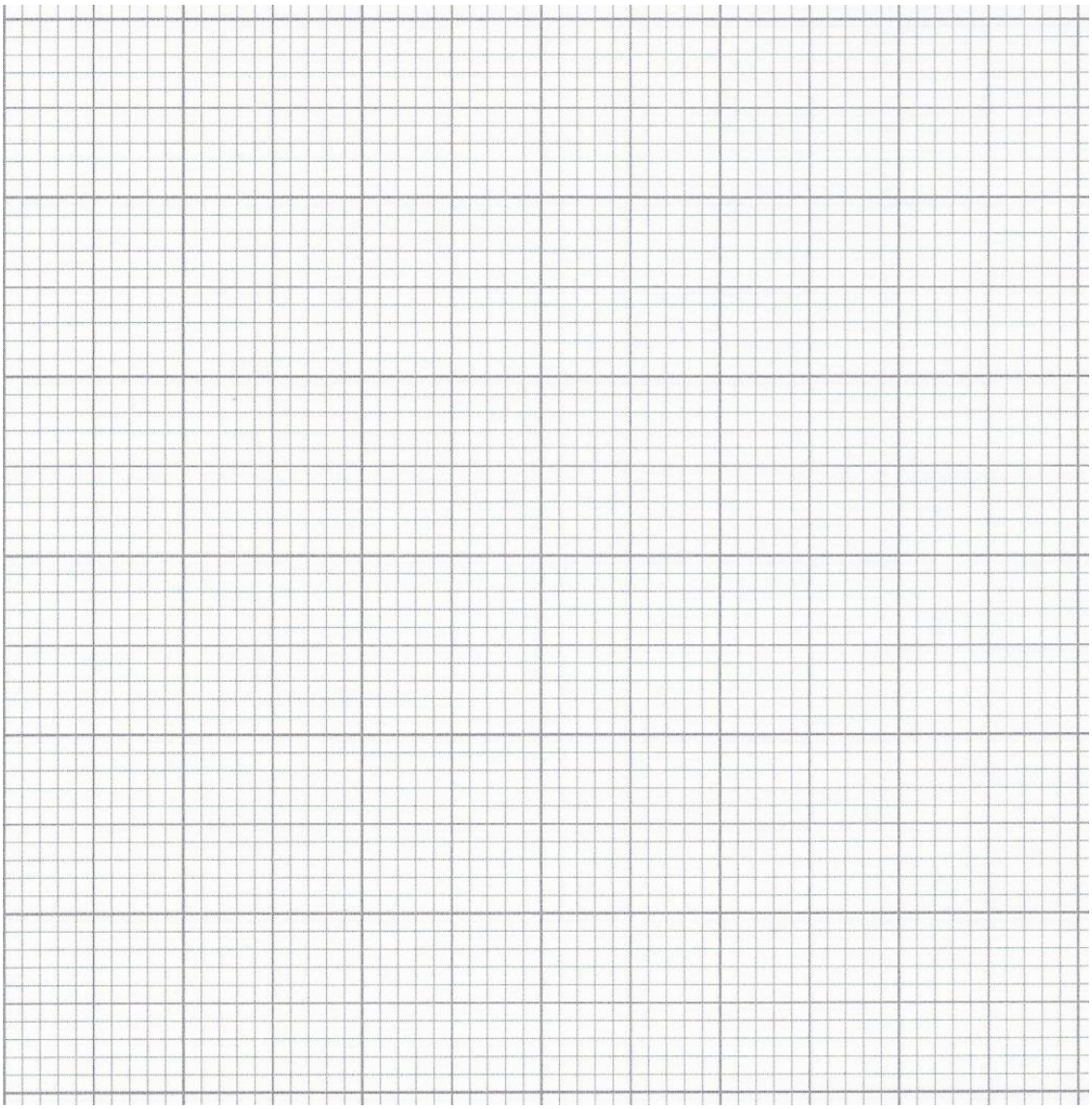


- 4 trampolines drawn to scale (3 cm × 1.8 cm = 3m × 1.8m) in Zone 3 [3 marks]
- Trampolines positioned symmetrically with 2 lines of symmetry shown [2 marks]
- Key/label for trampolines clearly indicated [1 mark]

(6 marks)

[6 marks]

Spare graph paper for Question 15

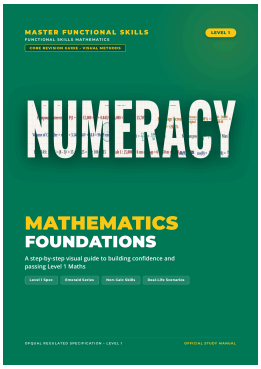


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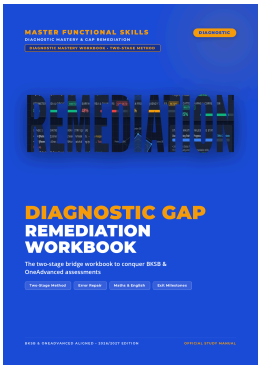


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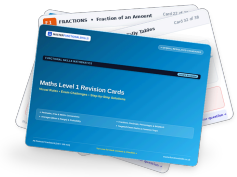


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